Question 1 (20 Marks) **Select the best correct answer and write it in the table below.**

1. The fuse symbol shown in the figure below is for:
    - A. IEEE standard
    - B. ANSI standard
    - C. IEEE and ANSI standard
    - D. Old symbol
2. During the normal operation of the circuit, the fuse wire is:
    - A. A very low resistance component
    - B. A very high resistance component
    - C. A very high inductive component
    - D. A very low inductive component
3. The amount of current that a fuse can easily conduct without interrupting the circuit is known as:
    - A. Breaking capacity
    - B. I^2T value
    - C. Response characteristics
    - D. Current carrying capacity
4. The electronic fuse is a type of:
    - A. Surface Mount Technology (SMT) fuse
    - B. Bolted type fuse
    - C. High Rupturing Capacity (HRC) fuse
    - D. Rewirable type fuse
5. When a fault occurs in the system, the type of fuse that automatically resets after operation is classified as:
    - A. Current limiting fuse
    - B. Current limiting fuse
    - C. Reusable fuse
    - D. One-time-only fuse
6. The type of fuse used to protect transformers as an expulsion type fuse is:
    - A. Reusable fuse type
    - B. Dropout fuse type
    - C. Fast-acting chip fuse type
    - D. High-current-rated chip fuse type
7. For an electrical load of **1760 W** connected to a **220 V** supply, the required fuse rating will be:
    - A. 10 A
    - B. 8 A
    - C. 12 A
    - D. 17.6 A
8. The smallest current that will cause the fuse element to melt is:
    - A. Maximum using current
    - B. Minimum fusing current
    - C. Current rating
    - D. Fusing factor
9. For circuit breakers, the length of the arc is directly proportional to:
    - A. Charge capacitor
    - B. Magnetic field
    - C. Voltage
    - D. Cross-section
10. For air circuit breakers under fault current:

- A. The main contacts open and the arcing contacts open.
- B. The main contacts close and the arcing contacts open.
- C. The main contacts open and the arcing contacts close.
- D. The main contacts close and the arcing contacts close.

Answer Key

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	**B**	**A**	**D**	**A**	**C**	**B**	**A**	**B**	**C**	**C**

Question 2 (5 Marks)

Mention the classification of LV fuses depending on the fusing factor.

Answer

Fuse Type	Fusing Factor (FF)
FF Type	($1.75 \leq FF \leq 2.5$)
Q2 Type	($1.50 \leq FF \leq 1.75$)
Q1 Type	($1.25 \leq FF \leq 1.50$)
P Type	($FF \leq 1.25$)

Question 3 (5 Marks)

Mention five types of Low-Voltage AC circuit breakers.

Answer

1. **MCCB** – Molded Case Circuit Breaker
2. **MCB** – Miniature Circuit Breaker
3. **RCCB** – Residual Current Circuit Breaker
4. **GFCI** – Ground Fault Circuit Interrupter
5. **ELCB** – Earth Leakage Circuit Breaker